

Glucomin 850mg Tablet

DESCRIPTION:

A white to off white oblong shaped film coated tablet.

COMPOSITION:

Each film coated tablet contains 850mg Metformin HCl equivalent to Metformin.

PHARMACODYNAMIC:

Metformin is a biguanide derivative of guanidine, used for treating type II diabetes mellitus. It has antihyperglycaemic effects, which lowers both basal and postprandial plasma glucose. It does not increase insulin secretion and hence, does not cause hypoglycaemia.

Metformin acts via 3 mechanisms:

- reduction in hepatic glucose production,
- reduction in intestinal glucose absorption, and
- Increased insulin sensitivity (improved peripheral glucose uptake and utilization).

Metformin stimulates intracellular glycogen synthesis by acting on glycogen synthase. Metformin increases the transport capacity of all types of membrane glucose transporters (GLUTs). In humans, independently of its action on glycaemia, metformin has favourable effects on lipid metabolism.

PHARMACOKINETICS:

Absorption:

After an oral dose of metformin hydrochloride tablet, maximum plasma concentration (C_{max}) is reached in approximately 2.5 hours (t_{max}).

Metformin hydrochloride is slowly and incompletely absorbed from the gastrointestinal tract.

It is assumed that the pharmacokinetics of metformin absorption is non-linear.

Bioavailability for oral metformin is approximately 50-60% in healthy subjects.

Dose proportionally is lacking due to decreasing absorption with increasing doses.

Steady state plasma concentrations for metformin are reached within 24 to 48 hours and are generally less than 1 microgram/ml at recommended doses and dosing schedules.

Food decreases the extent and delays the time to achieve maximum absorption. The plasma peak concentration may be 40% lower with food administration.

Distribution:

Plasma protein-binding is negligible.

Metformin crosses the placenta and is distributed into breast milk in small amounts.

Metformin partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean volume of distribution (V_d) ranged between 62-276L.

Metabolism:

Metformin is excreted unchanged in the urine. No metabolites are identified in humans.

Elimination:

Renal clearance of metformin is 450 to 540 ml/min indicating that metformin is eliminated by glomerular filtration and tubular secretion.

The plasma elimination half-life is approximately 2 to 6 hours.

When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged leading to increased levels of metformin in plasma.

Hemodialysis effectively removes metformin and corrects metabolic acidosis induced by metformin.

INDICATIONS:

Treatment of type 2 diabetes mellitus, when dietary management and exercise alone does not result in adequate glycemic control.

- In Adults, metformin may be used as monotherapy or in combination with other oral anti-diabetic agents or with insulin.
- In children from 10 years of age and adolescents, metformin may be used as monotherapy or in combination with insulin.

A reduction of diabetic complication has been shown in overweight type 2 diabetic adult patients treated with metformin as first-line therapy after diet failure.

In Type 1 diabetes, metformin may be given as an adjuvant to patients whose diabetic are poorly controlled.

DOSAGE & ADMINISTRATION:

Adults

Monotherapy and combination with other oral antidiabetic agents

- The usual of starting dose is one tablet, 2 or 3 times daily given during or after meals.
- After 10 to 15 days, the dose should be adjusted on the basis of blood glucose measurement. A slow increase of dose may improve gastrointestinal tolerability.
- The maximum recommended dose of metformin is 3 g daily.
- If transfer from another oral antidiabetic agent is intended: discontinue the other agent and initiate metformin at the dose indicate above.

Combination with insulin

Metformin and insulin may be used in combination therapy to achieve better blood glucose control combination therapy of. When insulin dosage is adjusted on the basis of blood glucose measurements, metformin is given at the usual starting dose of one tablet 2-3 times daily.

Children and adolescents

Monotherapy and combination with insulin

- Glucomin 850 can be used in children from 10 years of age and adolescent.
The usual starting dose 850mg once daily once daily, given during meals or after meals.
- After 10 to 15 days, the dose should be adjusted on the basis of blood glucose measurements.
A slow increase in dose may improve gastrointestinal tolerability. The maximum recommended dose of metformin is 2g daily. It taken as 2 or 3 divided doses.

Renal impairment

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

GFR mL/min	Total maximum daily dose (to be divided into 2-3 daily doses)	Additional considerations
60-89	3000 mg	Dose reduction may be considered in relation to declining renal function.
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of

30-44	1000 mg	metformin. The starting dose is at most half of the maximum dose.
<30	-	Metformin is contraindicated.

ROUTE OF ADMINISTRATION:

Oral administration

CONTRAINDICATIONS:

- Hypersensitivity to metformin or to any of the excipients listed in section 6.1.
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis).
- Diabetic pre-coma.
- Severely reduced kidney function (GFR < 30 mL/min).
- Acute conditions with the potential to alter renal function such as: dehydration, severe infection, shock.
- Disease which may cause tissue hypoxia (especially acute disease, or worsening of chronic disease) such as: decompensated heart failure, respiratory failure, recent myocardial infarction, shock.
- Hepatic insufficiency, acute alcohol intoxication, alcoholism.

WARNINGS AND PRECAUTIONS:

Lactic acidosis:

Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (< 7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal function:

GFR should be assessed before treatment initiation and regularly thereafter, see Dosage & Administration. Metformin is contraindicated in patients with GFR<30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function, see Contraindications.

Metformin may reduce vitamin B 12 serum levels. The risk of low vitamin B 12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B 12 monitoring could be necessary in patients with risk factors for vitamin B 12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contraindicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

Surgery:

Metformin must be discontinued 48 hours before elective surgery under general, spinal or peridural anesthesia. Therapy may be restarted no earlier than 48 hours following surgery or resumption of oral nutrition and only if normal renal function has been established.

Administration of iodinated contrast media:

The intravascular administration of iodinated contrast media in radiologic studies can lead to possible acute alteration of renal function resulting in increased risk of lactic acidosis. Metformin must be discontinued prior to, or at the time of the test and not be reinstituted until 48 hours afterwards, and only after renal function has been re-evaluated and found to be normal.

Children and adolescents:

The diagnosis of type 2 diabetes mellitus should be confirmed before treatment with metformin is initiated. Careful follow-up, especially in prepubescent children, is recommended.

Particular caution is recommended when prescribing to children aged between 10 and 12 years.

Elderly:

The metformin dosage should be adjusted based on renal function. Careful monitoring of renal function is required for this group of patients.

Patients should be cautioned against excessive alcohol intake, either acute or chronic, when taking metformin hydrochloride, since alcohol potentiates the effects of metformin hydrochloride on lactate metabolism.

Other precautions:

All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.

The usual laboratory tests for diabetes monitoring should be performed regularly. Metformin alone does not cause hypoglycaemia, but caution is advised when it is used concurrently with insulin or other oral antidiabetics (e.g. sulfonylureas or meglitinides).

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

Metformin monotherapy does not cause hypoglycaemia and therefore has no effect on the ability to drive or to use machines. However, patients should be alerted to the risk of hypoglycaemia when metformin is used in combination with other antidiabetic agents (e.g. sulfonylureas, insulin or meglitinides).

DRUG INTERACTIONS:

Concomitant use not recommended

Alcohol

Alcohol intoxication is associated with an increased risk of lactic acidosis, particularly in case of fasting, malnutrition or hepatic impairment.

Iodinated contrast agents

Metformin must be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable.

Combinations requiring precautions for use

Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclo-oxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.

Medicinal products with intrinsic hyperglycaemic activity (e.g. glucocorticoids (systemic and local routes) and sympathomimetics)

More frequent blood glucose monitoring may be required, especially at the beginning of treatment. If necessary, adjust the metformin dosage during therapy with the respective medicinal product and upon its discontinuation.

Organic cation transporters (OCT)

Metformin is a substrate of both transporters OCT1 and OCT2.

Co-administration of metformin with

- Inhibitors of OCT1 (such as verapamil) may reduce efficacy of metformin.
- Inducers of OCT1 (such as rifampicin) may increase gastrointestinal absorption and efficacy of metformin.
- Inhibitors of OCT2 (such as cimetidine, dolutegravir, ranolazine, trimethoprim, vandetanib, isavuconazole) may decrease the renal elimination of metformin and thus lead to an increase in metformin plasma concentration.
- Inhibitors of both OCT1 and OCT2 (such as crizotinib, olaparib) may alter efficacy and renal elimination of metformin.

Caution is therefore advised, especially in patients with renal impairment, when these drugs are co-administered with metformin, as metformin plasma concentration may increase. If needed, dose adjustment of metformin may be considered as OCT inhibitors/inducers may alter the efficacy of metformin.

PREGNANCY & LACTATION:

Pregnancy

Uncontrolled diabetes during pregnancy (gestational or permanent) is associated with increased risk of congenital abnormalities and perinatal mortality. There are limited data to suggest that metformin does not increase the risk of congenital abnormalities and does not adversely affect pregnancy outcome in diabetic women. Insulin is generally preferred for treatment of diabetes in women planning on becoming pregnant and during pregnancy to maintain blood glucose levels as close to normal as possible, to reduce the risk of malformations of the fetus.

Lactation

Metformin may be distributed into breast milk, and that the possible effects on the infant should be considered if women wish to breastfeed while receiving the drug.

Since only limited data are available, breast-feeding is not recommended during metformin treatment. A decision on whether to discontinue breast-feeding should be made, taking into account the benefit of breastfeeding and the potential risk to adverse effects on the child.

Fertility

Fertility of male or female rats was unaffected by metformin when administered at doses as high as 600 mg/kg/day, which is approximately three times the maximum recommended human daily dose based on body surface area comparisons.

SIDE EFFECTS:

During treatment initiation, the most common adverse reactions are nausea, vomiting, diarrhoea, abdominal pain and loss of appetite which resolve spontaneously in most cases. To prevent them, it is recommended to take metformin in 2 or 3 daily doses and to increase slowly the doses.

The following adverse reactions may occur under treatment with metformin.

Metabolism and nutrition disorders

Very rare

- Lactic acidosis (see section 4.4).
 - Decrease of vitamin B12 absorption with decrease of serum levels during long-term use of metformin.
- Consideration of such aetiology is recommended if a patient presents with megaloblastic anaemia.

Nervous system disorders

Common

- Taste disturbance

Gastrointestinal disorders

Very common

- Gastrointestinal disorders such as nausea, vomiting, diarrhoea, abdominal pain and loss of appetite. These undesirable effects occur most frequently during initiation of therapy and resolve spontaneously in most cases. To prevent them, it is recommended that metformin be taken in 2 or 3 daily doses during or after meals. A slow increase of the dose may also improve gastrointestinal tolerability.

Hepatobiliary disorders

Very rare

- Isolated reports of liver function tests abnormalities or hepatitis resolving upon metformin discontinuation.

Skin and subcutaneous tissue disorders

Very rare

- Skin reactions such as erythema, pruritus, urticaria

Pediatric population

In published and post marketing data and in controlled clinical studies in a limited pediatric population aged 10-16 years treated during 1 year, adverse event reporting was similar in nature and severity to that reported in adults.

Metabolism and nutrition disorders

Common: Vitamin B12 decrease/deficiency

OVERDOSAGE & TREATMENT:

Symptoms

Hypoglycaemia has not been seen with metformin hydrochloride doses of up to 85 g, although lactic acidosis has occurred in such circumstances. High overdose of metformin or concomitant risks may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital.

Treatment

Symptomatic and supportive care is the mainstay of treatment in patients who present with mild to moderate biguanide toxicity. Activated charcoal can be considered after large ingestions.

Hemodialysis is the most effective method in removing lactate and metformin.

PRESENTATION:

Blister of 10 tablets. Box of 1, 3, 10, 50 and 100 strips.

STORAGE CONDITION:

Store below 30°C

MANUFACTURER & PRODUCT REGISTRATION HOLDER:

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5 ½ Mile Off Jalan Meru,

41050 Klang,

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